Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year II Semester 2024-25

Assignment Sheet-I
Session 2024-25
(Even Semester)

Subject: Problem Solving using Object Oriented Paradigm (CSUL201) **Max Marks-40**

Due Date: 18/3/25

Roll No.	1-6	7-12	13-18	19-24	25-30	31-36	37-42	43-48	49-54	55-60	61-66
Set-Number	I	II	III	IV	V	VI	VII	VIII	IX	X	XI

(Set-I)

Q. No	Question	BL	CO	MM
1	What are the advantages of OOP over POP?	2	1	10
2	Explain encapsulation with an example.	2	1	10
3	Define a student class with attributes name, rollNumber, and marks. Implement methods to set and display student details.	3	2	10
4	Implement a Circle class with a constructor that initializes radius and a method to calculate area.	3	2	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year II Semester 2024-25

Assignment Sheet-I
Session 2024-25
(Even Semester)

Subject: Problem Solving using Object Oriented Paradigm (CSUL201) **Max Marks-40**

(Set-II)

Q. No	Question	BL	CO	MM
1	Difference between structures and classes with example.	2	1	10
2	Explain the difference between primitive and derived data types.	2	1	10
3	Implement a Person class with private attributes name and age. Use public methods to set and get values.	3	2	10
4	Create a Bank class with a static function to display the bank's interest rate.	3	2	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year II Semester 2024-25

Assignment Sheet-I
Session 2024-25
(Even Semester)

Subject: Problem Solving using Object Oriented Paradigm (CSUL201) **Max Marks-40**

(Set-III)

Q. No	Question	BL	CO	MM
1	What are the features of Object-Oriented Programming paradigm?	2	1	10
2	Explain with example “cin” and "cout".	2	1	10
3	Define a BankAccount class with attributes accountNumber, holderName, and balance. Implement methods for deposit and withdrawal.	3	2	10
4	Create a Point class with x and y coordinates. Implement a default constructor and a parameterized constructor.	3	2	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year I Semester 2024-25

Assignment Sheet-I
Session 2024-25
(Even Semester)

Subject: Problem Solving using Object Oriented Paradigm (CSUL201) **Max Marks-40**

(Set-IV)

Q. No	Question	BL	CO	MM
1	How does data abstraction help in object-oriented programming?	2	1	10
2	Compare function calling mechanisms in OOP and POP.	2	1	10
3	Create a Mobile class with multiple constructors, including a copy constructor.	3	2	10
4	Implement a Book class where the title attribute is private, and a member function updates it.	3	2	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year I Semester 2024-25

Assignment Sheet-I
Session 2024-25
(Even Semester)

Subject: Problem Solving using Object Oriented Paradigm (CSUL201) **Max Marks-40**

(Set-V)

Q. No	Question	BL	CO	MM
1	Explain how data hiding is implemented in classes with example.	2	1	10
2	Differentiate between structure and class with example.	2	1	10
3	Create a Circle class with an inline function to return the circumference.	3	2	10
4	Define a Distance class and write a function that takes two objects and returns the sum of distances.	3	2	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year I Semester 2024-25

Assignment Sheet-I
Session 2024-25
(Even Semester)

Subject: Problem Solving using Object Oriented Paradigm (CSUL201) **Max Marks-40**

(Set-VI)

Q. No	Question	BL	CO	MM
1	Compare the usability of structures and classes in real-world applications.	2	1	10
2	Explain the role of objects in OOP.	2	1	10
3	Design a Circle class with a radius attribute and member functions to calculate area and circumference.	3	2	10
4	Define a Laptop class with attributes like brand, processor, and RAM. Use a parameterized constructor to initialize attributes.	3	2	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year II Semester 2023-24

Assignment Sheet-I
Session 2024-25
(Even Semester)

Subject: Problem Solving using Object Oriented Paradigm (CSUL201) **Max Marks-40**

(Set-VII)

Q. No.	Question	BL	CO	MM
1	How does OOP differ from procedural programming in terms of code reusability?	2	1	10
2	How does function implementation differ in structures and classes?	2	1	10
3	Develop a ComplexNumber class with real and imaginary parts as attributes. Implement a method to add two complex numbers.	3	2	10
4	Design a Car class with a copy constructor to copy details from another object.	3	2	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year II Semester 2023-24

Assignment Sheet-I
Session 2024-25
(Even Semester)

Subject: Problem Solving using Object Oriented Paradigm (CSUL201) **Max Marks-40**

(Set-VIII)

Q. No.	Question	BL	CO	MM
1	Explain the class and objects.	2	1	10
2	Explain Input and output streams (cin, cout) with Example.	2	1	10
3	Create a Time class with hours, minutes, and seconds as attributes. Implement a method to add two Time objects.	3	2	10
4	Implement a Rectangle class with a parameterized constructor to initialize length and breadth.	3	2	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year II Semester 2023-24

Assignment Sheet-I
Session 2024-25
(Even Semester)

Subject: Problem Solving using Object Oriented Paradigm (CSUL201) **Max Marks-40**

(Set IX)

Q. No.	Question	BL	CO	MM
1	What is encapsulation. Explain with example.	2	1	10
2	Why is OOP considered better for large software development?	2	1	10
3	Create a Student class with a private member marks and a function to modify it using a friend method.	3	2	10
4	Implement a Company class with a static member function to print company policies.	3	2	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year II Semester 2023-24

Assignment Sheet-I
Session 2024-25
(Even Semester)

Subject: Problem Solving using Object Oriented Paradigm (CSUL201) Max Marks-40

(Set X)

Q. No.	Question	BL	CO	MM
1	In what ways does OOP enhance program efficiency.	2	1	10
2	Difference between OOP and POP.	2	1	10
3	Define two classes Car and Bike. Implement a friend function that compares their speeds.	3	2	10
4	Create a Shape class with protected attributes and implement a derived class Circle to access them.	3	2	10

Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

B. Tech. I Year II Semester 2023-24

Assignment Sheet-I
Session 2024-25
(Even Semester)

Subject: Problem Solving using Object Oriented Paradigm (CSUL201) **Max Marks-40**

(Set-XI)

Q. No.	Question	BL	CO	MM
1	Why is encapsulation important in OOP?	2	1	10
2	Define class and object with syntax.	2	1	10
3	Implement a Distance class with private attributes and use a friend function to add two distances.	3	2	10
4	Create a Book class with attributes title, author, and price. Write a program to create objects of this class and display book details.	3	2	10